

Economic Stress & Behaviour Change in Europe

A Time Series Analysis of Protest Activity and Social Anxiety (2010–2023)

Beatriz N. Coelho

Undergraduate Research Project

Bachelor's Degree in Sociology

Faculty of Economics, University of Coimbra (FEUC)

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Abstract

This study investigates the relationship between economic stress and behavioural outcomes — protest activity and social anxiety — across seven European countries from 2010 to 2023. A composite Economic Stress Index was constructed from unemployment, inflation, and GDP growth data sourced from Eurostat and the World Bank. Protest event data were compiled from ACLED public summaries, and social anxiety indicators were extracted from Google Trends using the pytrends API.

The analysis reveals strong positive correlations between economic stress and "economic crisis" search behaviour (+0.700) and moderate correlations with protest activity (+0.248) at the pooled level. Within-country analysis uncovers substantial heterogeneity: Greece (+0.728), Spain (+0.585), and Portugal (+0.428) show strong positive stress–protest associations, while France (−0.228) and Italy (−0.221) exhibit weak negative relationships. Granger causality tests do not find evidence that unemployment predicts anxiety at conventional significance levels, suggesting that economic stress and mental health indicators follow independent long-term trajectories.

The findings suggest that economic stress translates into collective action primarily in Southern European countries with weaker institutional buffers, while Northern and Central European welfare systems may decouple economic hardship from social mobilisation. The study contributes to computational social science by integrating macroeconomic indicators, protest event data, and digital behavioural traces into a unified analytical framework.

Keywords: economic stress, protests, social anxiety, Google Trends, computational social science, Europe, Granger causality

1. Introduction

The relationship between economic conditions and collective behaviour has preoccupied social scientists since Durkheim's foundational work on anomie. Economic crises — from the Great Depression to the 2008 financial crash and the COVID-19 pandemic — consistently generate not only material hardship but also measurable shifts in public mood and political behaviour. The Eurozone crisis of 2010–2013 produced mass protests across Southern Europe, while the COVID-19 pandemic triggered both economic contraction and a documented surge in anxiety and depression worldwide.

This study asks: To what extent do economic shocks predict changes in protest activity and public anxiety across European countries? Four hypotheses are tested:

H1: Rising unemployment is associated with increased protest activity.

H2: Inflation spikes correlate with heightened social anxiety.

H3: GDP contraction precedes peaks in protest frequency.

H4: The 2020 COVID-19 shock produced behavioural responses distinct from other recessions.

The study covers seven European countries — Portugal, Spain, France, Germany, Italy, the Netherlands, and Greece — over the period 2010 to 2023. A composite Economic Stress Index is constructed from unemployment, inflation, and GDP growth. This is then correlated with protest event data from ACLED and social anxiety indicators from Google Trends. Granger causality testing is applied to assess whether economic variables temporally precede behavioural outcomes.

2. Data and Methods

2.1 Data Sources

Three categories of data were collected.

Economic indicators: Unemployment rates were obtained from Eurostat (tipsun20 dataset). GDP growth and inflation rates were sourced from the World Bank (NY.GDP.MKTP.KD.ZG

and FP.CPI.TOTL.ZG indicators). Data were extracted for the period 2010–2023 at annual frequency.

Protest events: Data on protest counts were compiled from ACLED (Armed Conflict Location & Event Data) public summaries. As full ACLED access requires registration, the data used are curated approximations based on publicly available country-level summaries.

Social anxiety: Google Trends data were extracted using the pytrends Python library. Search volume indices for "anxiety," "economic crisis," and "depression" were retrieved at annual frequency for each country. Google Trends data are normalised on a 0–100 scale relative to the peak search volume within each country's time series.

2.2 Composite Economic Stress Index

A standardised composite index was constructed as:

$$\text{Stress Index} = Z(\text{unemployment}) + Z(\text{inflation}) - Z(\text{GDP growth})$$

This formulation ensures that higher values indicate greater economic distress: rising unemployment and inflation increase the index, while rising GDP growth decreases it.

2.3 Analytical Strategy

The analysis proceeds in four stages. First, descriptive statistics and time series visualisation establish the broad patterns. Second, Pearson correlation coefficients are computed at both the pooled level and within each country. Third, scatter plots and heatmaps visualise the bivariate relationships. Fourth, Granger causality tests assess whether lagged values of unemployment predict contemporaneous anxiety levels, using Portugal as a case study.

2.4 Limitations

Several limitations should be noted. The ACLED protest data are approximations based on public summaries, not exact event counts. Google Trends indices are relative, not absolute measures of search volume. The annual frequency reduces the sample to 14 observations per country, limiting the statistical power of time series tests. Correlation does not imply causation, and the relationships observed may be influenced by omitted variables such as political opportunity structures, media coverage, and cultural attitudes toward protest.

3. Results

3.1 Descriptive Statistics

The Economic Stress Index peaked during two periods across the sample: 2011–2013 (Eurozone crisis) and 2020 (COVID-19). Spain recorded the highest single-year stress value (2.86 in 2020), followed by Greece (2.31 in 2020). Germany and the Netherlands consistently registered below-average stress throughout the period.

3.2 Correlation Analysis

At the pooled level, the Economic Stress Index correlates strongly with "economic crisis" Google searches (+0.700), moderately with unemployment (+0.613) and protest events (+0.248), and weakly with anxiety (+0.141).

Within-country stress–protest correlations reveal a clear North–South divide:

Table 1: Within-Country Correlations (Stress Index vs Protest Events)

Country	Correlation
Greece	+0.728
Spain	+0.585
Portugal	+0.428
Germany	+0.076
France	−0.228
Italy	−0.221
Netherlands	−0.159

Southern European countries show the strongest positive associations between economic stress and protest activity. In contrast, France, Italy, and the Netherlands exhibit weak or negative correlations, suggesting that institutional factors mediate the translation of economic grievances into street mobilisation.

3.3 Granger Causality Test

Granger causality tests were applied to assess whether lagged unemployment values predict anxiety levels in Portugal. At both one and two lags, the null hypothesis of no Granger causality could not be rejected ($p > 0.05$). This suggests that unemployment and anxiety follow independent long-term trends rather than a short-term causal relationship. The rise in anxiety searches appears to be a secular trend across all countries, potentially driven by increased digitalisation and mental health awareness rather than economic cycles.

3.4 COVID-19 Effect

The 2020 COVID-19 shock produced the highest stress index values in most countries. However, protest activity declined in 2020 relative to 2019 in several countries (Portugal, Spain, Italy), likely due to lockdown restrictions limiting physical mobilisation. This suggests that the relationship between economic stress and protest behaviour is moderated by contextual factors such as public health measures and policing.

4. Discussion

The findings of this study paint a nuanced picture of the economic stress–behaviour nexus. At the aggregate level, the pooled correlation between stress and protest activity (+0.248) is positive but modest, and the correlation with anxiety (+0.141) is weak. These aggregate figures, however, mask substantial cross-national heterogeneity.

The North–South divide in stress–protest correlations is the study's most striking finding. In Greece, Spain, and Portugal — countries that experienced the most severe economic contractions during the Eurozone crisis and that have comparatively weaker social safety nets — economic stress translates into collective action. In France and Italy, protest traditions appear to operate independently of economic cycles, reflecting political cultures where street mobilisation is a routine form of political expression regardless of economic conditions. In

Germany and the Netherlands, robust welfare systems and institutionalised channels of interest representation may absorb economic grievances before they escalate into protest.

The absence of Granger causality between unemployment and anxiety is methodologically important. It cautions against simplistic narratives that economic hardship directly causes mental health deterioration. The secular rise in anxiety searches across all countries — from Portugal (21 in 2010 to 93 in 2022) to Germany (38 to 42) — points to structural factors beyond economic cycles: increased mental health awareness, reduced stigma around help-seeking, and the proliferation of digital devices enabling private health-related searches.

The COVID-19 analysis reveals an important moderator: physical possibility. The pandemic produced extreme economic stress but suppressed protest activity through lockdowns, illustrating that economic grievances require opportunity structures — freedom of assembly, reduced policing of public space — to translate into observable collective action.

5. Conclusion

This study demonstrates that the relationship between economic stress and behavioural outcomes is neither uniform nor deterministic. While economic hardship correlates with protest activity in Southern Europe, the relationship is weak or absent in countries with stronger institutional buffers. Anxiety, meanwhile, follows a secular upward trend largely decoupled from economic cycles.

The study's methodological contribution is threefold. First, it demonstrates the feasibility of integrating official statistics (Eurostat, World Bank) with digital behavioural traces (Google Trends) and conflict event data (ACLED) in an undergraduate research framework. Second, it illustrates the importance of disaggregated, within-country analysis to avoid Simpson's Paradox. Third, it introduces a composite Economic Stress Index that synthesises multiple macroeconomic indicators into a single interpretable metric.

From a policy perspective, the findings caution against economically deterministic approaches to social stability. Economic recovery alone may not reduce protest activity in countries where political grievances have autonomous drivers. Conversely, robust welfare systems appear to buffer the translation of economic hardship into social unrest, supporting the case for automatic stabilisers and social protection floors.

6. Future Research Directions

Several extensions of this work are planned or underway. First, registration with ACLED will enable the use of exact, georeferenced protest event data, improving measurement precision. Second, expanding the analysis to monthly frequency will increase statistical power for time series methods. Third, incorporating social media sentiment analysis (Twitter/X API) will provide more direct measures of public mood. Fourth, applying machine learning techniques for protest prediction could transform this descriptive framework into a forecasting tool. Fifth, building an Early Warning Index for social instability would have direct policy applications. Finally, expanding the geographical scope to Latin American and Asian countries would test the generalisability of the North–South institutional buffer hypothesis.

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